MIT 8103 Advanced Database Systems

Fitness and Workout Tracking · John Folarin · Student ID 301931978

Case study

My case study is Fitness and Workout Tracking. I designed the database for members tracking their workouts and coaches reviewing member training history, with separate tables named members, exercises, workouts and workout_sets. It supports these operations: create and complete workout sessions; record exercise sets with repetitions and load; review workout history and training totals.

Portfolio 1: Database Design and Modelling

I organised the relational design into 4 tables. Primary keys identify records, foreign keys preserve the links, and checks enforce the stated business rules. At most one workout per member may have status in_progress; enforce with a partial unique constraint on member_id for in-progress workouts. The normalisation examples show the sample data at 1NF, 2NF and 3NF. All 32 deliberately invalid inserts were rejected without changing the valid sample rows.

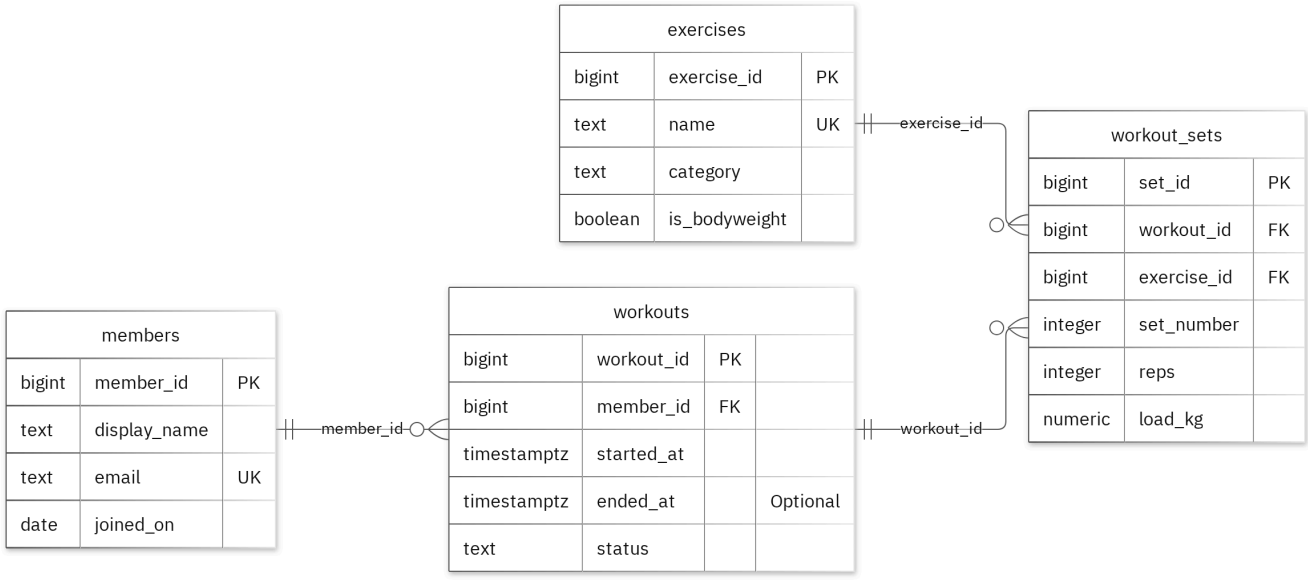


Figure 1. Tables, keys and relationships. PK: primary key; FK: foreign key; UK: unique value.

Portfolio 2: Query Processing and Optimisation

I ran three SQL queries against the sample records. The join returned 2 rows and included Avery Chen with Squat. For the index comparison, I used 50,000 separate performance rows. The median lookup time changed from 2.738 ms to 0.363 ms. Both phases returned the same records. The measured plan used the new index. These timings apply to this lookup and environment.

Portfolio 3: Transactions and Concurrency

I used a temporary counter on workouts to test COMMIT, ROLLBACK and READ COMMITTED isolation. The uncommitted update remained hidden from the other session, and ROLLBACK restored the earlier committed value. Without locking, two increments left the counter at 2. With SELECT FOR UPDATE, Session B waited for Session A and the final value was 3. The helper is confined to this exercise; the case's original business fields stay unchanged.

Portfolio 4: NoSQL and Advanced Data Models

I used training_events for the case's event documents. MongoDB inserted 4 sample events and returned the expected results for 4 queries. Validation rejected an invalid date type and a duplicate event ID. Embedded snapshots keep related details with an event, while the details field can differ by event type. These are fictional events; PostgreSQL continues to enforce its relational rules.

Repository and evidence

https://github.com/Folabamijoh-code/Folarin_John_301931978

The portfolio folders contain the SQL and JavaScript, explanations, query plans, result tables and screenshots.

Portfolio 5: Distributed and Cloud Database Exercise

I proposed three MongoDB replica members across separate failure zones. The discussion covers primary elections, majority acknowledgement, secondary lag and sharding trade-offs for training_events. Replication supports availability, while backups address recovery. PostgreSQL needs its own availability plan. This is the permitted discussion option; no live replica set or failover test was deployed.

Individual reflection

I kept members, exercises, workouts, workout_sets in separate tables so their details could be maintained without repeating them in every related record. At most one workout per member may have status in_progress; enforce with a partial unique constraint on member_id for in-progress workouts. The constraint tests check this alongside the required values and record links. One issue illustrated by the two-session exercise is that both sessions can read the same value before either update is committed. Locking the row before reading preserves both increments. The index comparison also shows why I need to check the execution plan as well as the final answer: the same result can require different amounts of database work.

AI-use declaration

I used GitHub Copilot and the portfolio generator to help prepare the database model, scripts and explanations. The generator ran the PostgreSQL and MongoDB exercises, checked their outputs and captured the result images. I remain responsible for checking and explaining the submitted work.

References

- PostgreSQL: constraints
- PostgreSQL: EXPLAIN
- PostgreSQL: isolation
- PostgreSQL: row locks
- MongoDB: data modelling
- MongoDB: replication
- MongoDB: sharding
- Mermaid: ER diagrams